MIL-T-49066A(EL)

10 January 1977

Superseding
MIL-T-49066(EL)
15 May 1975

MILITARY SPECIFICATION

Ⓐ TEST SET, TSEC/ST-23( )

This specification is approved for use by the Electronics Command Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for Test Set, TSEC/ST-23( ) which is used to test and maintain the HYL-3( )/TSEC, HYL-4( )/TSEC and MX-9331( )/URC. (See 6.1 and 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitiation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallet Material Handling, Wood, Stringer Construction 2 Way and 4 Way (Partial)

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood Wirebound
PPP-B-601	Box, Wood Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification for
MIL-M-38510	Microcircuits, General Specification for

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STANDARDS

Federal

FED-STD-356	Commerical Packaging of Supplies and Equipment

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1562	Lists of Standard Microcircuits

National Security Agency

NSA-2( )	Nameplates and marking of Electronic and Electromechanical Equipment

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DRAWINGS

Electronics Command

SM-B-662417	Specification, Electrical, 1A1A1 Assembly
SM-B-662418	Specification, Electrical, 1A1A2 Assembly
SM-B-662419	Specification, Electrical, 1A1A3 Assembly
SM-B-662420	Specification, Electrical, Test Set, TSEC/ST-23
DL-SM-B-662454	Test Set, TSEC/ST-23

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification, and of Drawing and Data List DL-SM-B-662454, including the requirements for parts, materials, and processes thereon.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the full requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4)

3.3.1 Microelectronic integrated circuits. (See 4.4) All microelectronic integrated circuit devices used in this equipment shall meet the requirements of MIL-M-38510 for Class B devices and their individual drawing requirements. Microcircuits shall be selected in accordance with MIL-STD-1562.

3.4 Finish, protective. The equipment shall be given a protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch up after mounting. The final paint film on Type I surfaces shall be final film E, semigloss, light-gray enamel, conforming to Specification MIL-F-14072. (See 4.4)

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3.5 Marking. (See 4.4)

3.5.1 Equipment and special warning plates. Equipment and special warning plates shall be in accordance with NSA-2( ). Equipment shall be serial numbered sequentially as specified in contract.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6 Service conditions. The equipment shall meet the following service conditions:

3.6.1 Vibration. (See 4.7.1) The equipment shall meet the requirements of 3.6.7 after being tested in accordance with 4.7.1. There shall be no evidence of electrical or mechanical degradation.

3.6.2 Shock bench handling. (See 4.7.2) The equipment shall meet the requirements of 3.6.7 after being tested in accordance with 4.7.2. There shall be no evidence of electrical or mechanical degradation.

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3.6.3 Humidity. (See 4.7.3) The equipment shall meet the requirements of 3.6.7 after being tested in accordance with 4.7.3. There shall be no evidence of corrosion, rust, deterioration of finishes, or other forms of degradation of the equipment as a result of this test.

3.6.4 Temperature. There shall be no evidence of electrical or mechanical degradation of the equipment when it is subjected to the following low and high temperature requirements.

3.6.4.1 Low temperature. (See 4.7.4). The equipment shall meet the requirements of 3.6.7 during and after being tested in accordance with 4.7.4. The lowest operating temperature shall be 0°C (+32°F), and the storage temperature shall be -57°C (-70°F).

3.6.4.2 High temperature. (See 4.7.5) The equipment shall meet the requirements of 3.6.7 during and after being tested in accordance with 4.7.5. The operating temperature shall be 52°C (125°F) without solar radiation. The non-operating temperature shall be 71°C (160°F).

3.6.5 Fungus. (See 4.7.6) After the fungus test of 4.7.6, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.6) as evidenced by growth colonization (which includes branching and sporulation) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instance of partial tubular germination shall not be included in this evaluation. Operation of the equipment shall not be required.

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3.6.6 Altitude. (See 4.7.7) The equipment shall meet the requirements of 3.6.7 when subjected to the test of 4.7.7.

3.6.7 Operational inspection. The operational test specified in 4.14.1 shall be performed to determine that the equipment is operable prior to packaging, and at such other times as required.

3.7 Electromagnetic interference (EMI). (See 4.8) The equipment shall meet the following requirements of MIL-STD-461, Notice 4:

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	RS03
CE04	CS06		

3.7.1 Bonding and grounding. Bonding for equipment current path returns, rf potentials, and shock hazard shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate areas of contact. The equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The dc resistance of bonds and grounds shall not exceed 2.5 milliohms.

3.8 Preconditioning.

3.8.1 Bounce preconditioning. (See 4.6) The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

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3.8.2 Electrical preconditioning/burn-in. (See 4.9) Prior to submission to the specified quality conformance inspections, each equipment shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free.

3.9 Interchangeability. (See 4.10) Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.10 Reliability. (See 4.11) The reliability of the equipment shall provide a specified (θ) mean-time-between-failure (MTBF) of 5000 hours. The required reliability shall be demonstrated in accordance with 4.11 of this specification.

3.11 Electrical performance.

3.11.1 Test Set, TSEC/ST-23( ). (See 4.14.1) The Test Setm TSEC/ST-23( )shall provide dc power, digital and analog test signals and the necessary switching functions to permit maintenance and repair of an HYL-3( )/TSEC, HYL-4( )/TSEC, MX-9331( )/URC, U-383( )/VR Power Converter and MX-9545/VR Vehicualr Power Adapter. The Test Set, TSEC/ST-23() shall have a switch on the front panel to select either the MX-9331( )/URC family or the HYL-3( )/TSEC family.

3.11.2 Digital mode. (See 4.14.2) The Test Set, TSEC/ST-23( ) shall exercise all the digital circuits of the HYL-3( )/TSEC, HYL-4( )/TSEC (18.75 Kbd rate); and the MX-9331( )/URC (16.0/18.75 Kbd rate) and determine on a go-no-go basis the operability of the respective devices in the digital mode.

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3.11.3 Analog mode. (See 4.14.3) The Test Set, TSEC/ST-23( ) shall fully exercise all analog circuits of the HYL-3( )/TSEC, HYL-4( )/TSEC and the MX-9331( )/URC, and determine on a go-no-go basis the operability of the respective devices in the analog mode.

3.11.4 Circuit card assembly, 1A1A1. (See 4.14.4) The circuit card assembly, 1A1A1 shall meet the performance requirements of drawing SM-B-662417.

3.11.5 Circuit card assembly, 1A1A2. (See 4.14.5) The circuit card assembly, 1A1A2 shall meet the performance requirements of drawing SM-B-662418.

3.11.6 Circuit card assembly, 1A1A3. (See 4.14.6) The circuit card assembly 1A1A3 shall meet the performance requirements of drawing SM-B-662419.

3.12 Systems safety engineering. (See 4.13)

3.12.1 Personnel hazards. Personnel Hazards shall be kept to a minimum through compliance with paragraph entitled "safety" of MIL-T-28800. Compliance with these requirements will be verified through a visual inspection.

3.12.2 Edge rounding. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

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3.12.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.13 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with requirements of MIL-P-11268 and requirement 9 of MIL-STD-454. (See 4.4)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. Does not include packaging. (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.12)

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4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 9 each first article units of the complete Test Set, TSEC/ST-23( ).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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Table I - First Article Inspection

Inspections a/	Req. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5-9
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units				
2. Bounce preconditioning	3.8.1	4.6	Inspections to be performed on all units.				
3. Electrical preconditioning	3.8.2	4.9	Inspections to be performed on all units.				
4. Group A inspection	See Table II		Inspections to be performed on all units.				
5. Group B inspection	See Table III		Inspections to be performed on all units.				
6. Group C inspection b/, c/							
Vibration	3.6.1	4.7.1	1				
Shock, bench handling	3.6.2	4.7.2			1		
Humidity	3.6.3	4.7.3		2			
Low temperature	3.6.4.1	4.7.4	2				
High temperature	3.6.4.2	4.7.5	3				
Fungus d/	3.6.5	4.7.6				1	
Altitude	3.6.6	4.7.7			2		
EMI	3.7	4.8			3		
7. Group D inspection							e/
Reliability	3.10	4.11					

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a/ The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b/ The implementation of inspections covered by 6, and 7 shall be expedited to satisfy contract delivery requirements.

c/ The numbers in the unit columns in the table covered by 6, specify the order of inspections for the indicated unit.

d/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after the fungus test before proceeding with subsequent inspection.

e/ Units 5-9: These units are to undergo inspections 1, 2, 3, 4, 5 and 7.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes; general	3.3
Microelectronic integrated circuits	3.3.1
Finish, protective	3.4
Marking	3.5
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.6, 4.9, and 4.5.1 through 4.5.5 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Subassemblies and completed equipments shall be inspected for conformance to the inspections specified in Table II. Each subassembly, after 100% inspection, shall be formed into discrete lots for quality assurance verification. The equipments assembled from lots accepted under subassembly verification shall be inspected 100% and formed into discrete lots for final QA verification. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification).

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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Table II - Group A Inspection

Inspection	Req. Para.	Test Para.	Major	Minor
Visual and Mechanical	3.13	4.13		
Circuit Card Assembly 1A1A1			1.0%	4.0%
Circuit Card Assembly 1A1A2			1.0%	4.0%
Circuit Card Assembly 1A1A3			1.0%	4.0%
Main Unit Subassembly TSEC/ST-23( )			1.0%	4.0%
Case, Combination			1.0%	4.0%
Cable Assembly, 115 VAC, Power			1.0%	4.0%
Cable Assembly, Special Purpose, Electrical CX-12452/U			1.0%	4.0%
Cable Assembly, Special Purpose, Electrical CX-12453/U			1.0%	4.0%
Electrical				
Operational inspection (Test Set, TSEC/ST-23( )	3.11.1	4.14.1	1.0%	A/
Subassemblies				
Circuit Card Assembly 1A1A	3.11.4	4.14.4	1.0%	
Circuit Card Assembly 1A1A2	3.11.5	4.14.5	1.0%	
Circuit Card Assembly 1A1A3	3.11.6	4.14.6	1.0%	
Digital Mode	3.11.2	4.14.2	1.0%	
Analog Mode	3.11.3	4.14.3	1.0%	

A/ Defects in all characteristics other than visual and mechanical are considered major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Table III - Group B Inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability Dimensional	3.9	4.10	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every month.

4.5.3.1.2 Subgroup 2. For this subgroup, one unit shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every two months.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month.

4.5.3.1.4 Subgroup 4. For this subgroup, one unit shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every six months.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in an order satisfactory to the Government.

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Table IV - Group C Inspection

Inspection	Req. Para.	Test Para.
Subgroup 1		
Low Temperature	3.6.4.1	4.7.4
High Temperature	3.6.4.2	4.7.5
Subgroup 2		
Humidity	3.6.3	4.7.3
Subgroup 3		
EMI	3.7	4.8
Altitude	3.6.6	4.7.7
Subgroup 4		
Fungus 1/	3.6.5	4.7.6
Shock, Bench Handling	3.6.2	4.7.2
Vibration	3.6.1	4.7.1

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.3 Group B and C failures. Actions required relative to Group B and C failures shall be as specified in the contract.

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and pass Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all damage.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.5.1 Sampling for Group D inspection. Samples shall be selected at random in accordance with the following:

Table V - Group D Inspection

Inspection	Req. Para.	Test Para.
Reliability	3.10	4.11

4.5.5.2 Group D failures. A failure occurrence during Group D inspection is defined as any deviation in performance from the requirements specified elsewhere in this specification.

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4.5.5.3 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.5.4 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and pass Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. (See 3.8.1) The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit of each lot of equipments shall be subjected to bounce preconditioning for 1 minute, prior to selection of units for inspection. After bounce preconditioning, the units shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.7 Service condition test. The following tests shall be performed on the equipment.

4.7.1 Vibration. (See 3.6.1) The equipment shall be subjected to sinusoidal vibration having a double amplitude of 0.015 inch, the frequency being varied uniformly between the limits of 10 Hz and 55 Hz. The rate of frequency change shall be linear and shall be such that a complete cycle (10-55-10) shall be traversed in 30 minutes. One cycle shall be performed along each of the three major axes of the equipment for a total test time of 90 minutes. At the completion of the test, the equipment shall meet specified performance requirements and have no physical damage.

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4.7.2 Shock, bench handling. (See 3.6.2) The equipment shall be tested in accordance with Method 516.2, Procedure V, of MIL-STD-810.

4.7.3 Humidity. (See 3.6.3) The equipment shall be tested in accordance with Method 507.1, Procedure III, of MIL-STD-810. Measurements shall be taken during the last 5 hours of Step 6 in each cycle.

4.7.4 Low temperature. (See 3.6.4.1) The equipment shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -57°C (-70°F) and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be 0°C (+32°F).

4.7.5 High temperature. (See 3.6.4.2) The equipment shall be tested in accordance with Method 501.1, Procedure II of MIL-STD-810. Step 4 shall be +71°C (+160°F) and Step 7 shall be 52°C (125°F).

4.7.6 Fungus. (See 3.6.5) The equipment shall be tested in accordance with Method 508.1, Procedure I of MIL-STD-810. The chassis shall be removed from its case and opened, and shall then be completely sprayed with the spore suspension and exposed to the ambient atmosphere. Operation of the equipment shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus or corrosion of any equipment surfaces.

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4.7.7 Altitude. (See 3.6.6) The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810.

4.8 Electromagnetic interference. (See 3.7) The equipment shall be tested to verify compliance with the requirements of 3.7. The test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, government approved, EMI test plan. Prior to the start of the EMI tests, compliance with the bonding requirement of 3.7.1 shall be verified. Bonding data shall be included in the EMI test report.

4.9 Electrical preconditioning (burn-in debugging). (See 3.8.2) All units submitted for Government acceptance shall be subjected to and pass a burn-in period. The burn-in period will consist of 48 hours of "ON" time with the last 24 hours of "ON" time failure free. Equipments shall be operated in the same manner and under the same environmental conditions as during reliability testing. If a unit fails testing during the final 24 hours of "ON" time it shall be repaired and returned to testing until it has accumulated 24 hours of failure free "ON" time. A failure shall be defined as the inability of the equipment to meet any of the performance requirements of Group A inspection. Failures shall be recorded and analyzed, but neither burn-in time nor burn-in failures shall be used to calculate equipment MTBF. Burn-in testing shall be performed prior to Group A testing. The equipment, including plug-in assemblies and spares shall have no less than 48 hours nor more than 144 hours of burn-in (exclusive of production test time).

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4.10 Inspection for dimensional interchangeability. (See 3.9) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.11 Reliability. (See 3.10) The following reliability test shall be performed on the equipment:

4.11.1 First article reliability test. The First Article Reliability Inspection Testing shall be in accordance with Test Plan V of MIL-STD-781 (See Table I).

4.11.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan IVA of MIL-STD-781. A minimum of four (4) samples shall be selected at random from each monthly production lot and sujected to the reliability test.

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4.11.3 Reliability test conditions. The reliability test described above shall be conducted under the conditions prescribed by Test Level B of MIL-STD-781. Paragraph 4.2.2.4, Test Length and applicable portions of 5.4, Test Procedure of MIL-STD-781 shall apply. Proper instrumentation shall be provided to insure recognition of failure. A failure is defined as the inability of an equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure from performance as required by this specification and the contractor's approved test procedure. If a pattern failure occurs, then a reject decision has been reached and corrective action shall be accomplished as specified elsewhere in the contract. Paragraphs 5.4.8, Determination of Compliance; 5.5.1, Failure Categories; 5.6, Verifying Repair; 5.8, Preventive Maintenance; 5.9, Restoration of Failed Equipment of MIL-STD-781 shall apply.

4.12 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.13 Visual and mechanical. (See 3.12 and 3.13) Equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.12 (Systems Safety Engineering) which can be determined visually.

4.14 Electrical test.

4.14.1 Operational inspection. (See 3.11.1) The Test Set, TSEC/ST-23( ) shall be tested in accordance with ECOM Drawing SM-B-662420 and meet the performance requirements therein.

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4.14.2 Digital mode. (See 3.11.2) All digital outputs shall be measured and tested to determine their suitability of fully exercising all digital circuits contained within the HYL-3( )/TSEC, HYL-4( )/TSEC and MX-9331( )/URC.

4.14.3 Analog mode. (See 3.11.3) All analog outputs shall be measured and tested to determine their suitability of fully exercising all analog circuits contained within the HYL-3( )/TSEC, HYL-4( )/TSEC and MX-9331( )/URC.

4.14.4 Circuit Card Assembly 1A1A1. (See 3.11.4) The Circuit Card Assembly 1A1A1 shall be tested in accordance with ECOM Drawing SM-B-662417.

4.14.5 Circuit Card Assembly 1A1A2. (See 3.11.5) The Circuit Card Assembly 1A1A2 shall be tested in accordance with ECOM Drawing SM-B-662418.

4.14.6 Circuit Card Assembly 1A1A3. (See 3.11.6) The Circuit Card Assembly 1A1A3 shall be tested in accordance with ECOM Drawing SM-B-662419.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Test Set, TSEC/ST-23( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Test Set, TSEC/ST-23( ) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Test Set, TSEC/ST-23( ). Each test set shall be individually unit packed method III as follows: place the set and accessories within the carrying case. Close the case and secure the fastenings provided. Cushion the case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place each cushioned case within a close fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with appendix of the box specification.

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5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Test Set, TSEC/ST-23( ) unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type 1, Finish A.

5.2.2 Level B. A quantity of Test Set, TSEC/ST-23( ) unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, Class 2, Style E, or PPP-B-636, Type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure, sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type 1, finish A, or shrink film.

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5.2.3 Commerical packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. The Test Set, TSEC/ST-23( ) contains five power supplies: -24 VDC, +12.5 VDC, +12 VDC, +3.3 VDC, and -6.5 VDC. The test set is used to provide input signals and output loads for functional troubleshooting and board fault isolation

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required (see 1.1).

c. Level A, level B or Commercial preservation and packing (see 5.1).

d. First article required (see 3.1).

e. Marking and shipping of samples.

f. When rough handling test is required.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example TSEC/ST-23W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

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6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Characteristics of externally supplied AC power sources. The externally supplied alternating-current voltage source may have a maximum total harmonic content of 5 percent with phase and amplitude relationship of individual harmonic frequencies so that the crest factor shall fall within the range of 1.27 and 1.56. The total harmonic content is defined as the square root of the sum of the squares of the individual harmonic frequencies. Crest factor is defined as the ratio of the crest voltage to the effective voltage, and has a value of 1.414 for a true sine wave.

6.6 Definitions.

6.6.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.6.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

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6.6.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.6.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.6.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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Army-EL

Preparing Activity

Army-EL

Project No. 5810-A037

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